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SUPERSEDING
MIL-C-18843E
June 4, 1986

COMMERCIAL ITEM DESCRIPTION

CUTTING MACHINE, MOTOR DRIVEN, PORTABLE

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL-C-18843E, which is cancelled.

1. ABSTRACT

1.1 Abstract. This description covers a portable, electric motor driven, oxygen fuel, cutting machine for cutting straight lines, arcs, and circles. The machine shall be capable of utilizing either a thermal or plasma torch.

2. SALIENT CHARACTERISTICS.

2.1 Design. The machines shall consist of a motorized carriage capable of transporting a cutting torch and guided by a set of tracks for straight cutting or by a radius rod assembly for cutting circles. The machine shall be new and one of the manufacturer's current models and shall include all components. parts and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage or distortion shall be accessible for adjustment, replacement and repair.

2.2 Carriage assembly. The carriage assembly shall consist of a housing, drive motor, drive train, controls, and other features necessary to meet the requirements of this description. The carriage housing shall completely enclose the drive components. The carriage shall be mounted on 2 drive and 2 caster wheels. The caster wheels shall be free to pivot through an angle of not less than 180 degrees about a vertical axis and shall have a lock for straight line cutting. The housing shall provide for mounting of the radius rod assembly.

2.3 Drive motor. The drive motor shall be of sufficient size and horsepower to meet maximum operating conditions.

Beneficial comments (recommendations, additions, deletions) clarifications, etc., and any data which may improve this document should be sent to: Defense General Supply Center (DGSC), ATTN: DGSC-SS, 8000 Jefferson Davis Highway, Richmond, Virginia 23297-5678.

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2.4 Drive train. The drive train shall consist of driving wheels connected to the drive motor through a clutch. The clutch shall disengage the drive motor from the driving wheels.

2.5 Control. Controls shall be furnished to change direction of carriage travel, speed of carriage travel, and to stop carriage travel. The controls shall be mounted in the carriage housing in a position convenient to the operator. Speed adjustments shall be possible while the carriage is in motion.

2.6 Torch mounting mechanism. The torch mounting mechanism shall consist of a torch holder, arm, mounting bracket, mounting post, and necessary clamps. The torch mounting mechanism shall provide for locating and locking the torch in any angular position between 90 and 180 degrees from vertical. The arm that attaches to the carriage and supports the torch shall be of sufficient strength to provide the accuracy specified in 2.14 with the torch in any position. Lateral adjustment of the torch shall be not less than 8-3/4 inches. The torch shall have sufficient vertical adjustment to accommodate the cutting capacity specified in 2.14. The horizontal and vertical adjustments shall be accomplished by means of rack and pinion gears.

2.7 Cutting torch. The machine shall be furnished with an oxygen-fuel gas cutting torch which meets the requirements of UL 123. The torch shall be of the machine type with a rack for vertical adjustment, gas valves, and a cutting tip. Unless otherwise specified (see 6.1), the torch shall be designed for use with oxygen and acetylene gas.

2.8 Cutting tips. The tips shall be fabricated of copper or copper-tellurium and shall have a heat conductivity not less than 90 percent of pure electrolytic copper. The tips shall be permanently and legibly identified for size, The center cutting oxygen orifice shall provide a stream of oxygen which travels down the centerline of the preheat flame group and shall not deviate from the center line by more than 2 degrees. A set of tips capable of cutting low carbon steel from 1/16 to 6 inches thick shall be furnished. Additional tips shall be as specified (see 6.1).

2.9 Hoses and fittings. The machine shall be furnished with all hoses and fittings required for operation. All hoses furnished shall comply with RMA IP-7. The fittings attached to the hoses shall conform to CGA E-1, connector numbers 020, 021, 022, or 023, as applicable. The hose fittings shall provide sufficient sealing to prevent gas leakage at all operating pressures. The length of the hose connecting the oxygen and full gas source to the machine shall be not less than 25 feet in length. When the machine is furnished with a hose manifold, a oxygen and fuel gas hose of sufficient length to reach from the manifold to the cutting torch shall be furnished.

2.10 Track. A steel track not less than 60 inches in length shall be provided with each machine for straight line cutting. The track shall be of welded construction or shall be fabricated from a single piece of stock.

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A precision ground guide groove shall be provided in each rail of the track to provide guidance of the machine's movements and to assure conformance with the accuracy specified in 2.14. Additional lengths of track shall be interconnectable to the orginial track and each additional piece of track provided as specified (see .1).

2.11 Radius rod assembly. A radius rod assembly shall be furnished with the machine for circle and arc cutting within the capacities specified.

2.12 Electric power cord. A three conductor power cord not less than 10 feet in length shall be furnished. The power cord may be permanently attached to the machine or may be connected by a receptacle and plug.

2.13 Electrical system. Unless otherwise specified (see 6.1), the electrical system shall conform to NFPA 79. Unless otherwise specified (see 6.1), the machine shall be designed to operate from a 120 volt, 60 HZ, single phase power source.

2.14 Performance. The machine shall be capable of cutting SAE 1020 low carbon steel up to 6 inches thick. The machine shall be capable of cutting a straight line 60 inches in length and a circle 4 inches in diameter with a deviation of + or - .062 inches on the straight or circle cut. The machine shall have a cutting range from not greater than 4 inches per minute (IPM) to not less than 32 IPM.

2.15 Additional performance requirements. Performance features required in addition to, or in lieu of the requirements herein, shall be as specified by the procuring activity (see 6.1).

2.16 Standard equipment. All standard equipment normally provided with the manufacturer's commercial machine shall be furnished.

2.17 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.1). The equipment provided shall be functional without requiring modification of the equipment or machine.

2.18 Repair parts. Required repair parts shall be furnished as specified (see 6.1).

2.19 General requirements.

2.19.1 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in ASTM E 380 and ANSI/IEEE 286 and all other requirements of this commercial item description are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerance specified in the inch-pound units, a request shall be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

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2.19.2 Safety and health requirements. All safety devices normally incorporated in the machine by the manufacture for the protection of the operator shall be furnished. All parts, components, mechanisims, and assembles furnished on the machine. whether or not specifically required herein, shall conform with all requirements of OSHA 29 CFR 1910. Additional safety and health requirements shall be as specified (see 6.1)

2.19.3 Environmental protection. During the manufacture of the machine, materials hazardous to the ecological system as prohibited by the Federal state, or local statutes in effect on the date of the contract shall not be used or emitted.

2.19.4 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable All parts shall be manufactured to such standards as will permit replacement or adjustment without modification of parts or the machine.

2.19.5 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

2.19.6 Operating and maintenance instructions. Operating and maintenance instructions shall be provided with each machine furnished. The instructions shall include directions and information necessary for the safe operation and maintenance of the machine and a list of replacement parts with part numbers. The instructions shall be written in the English language.

2.19.7 Marking on instruments, controls panels, charts, and plates. All words on the machine including all instruments, control panels, charts, plates and related attachments shall be in the English language. Characters shall be marked in boldface on a contrasting background. All plates shall be corrosion resistant.

2.19.8 Data plates. All standard data plates normally provided with the manufacturer's standard commercial machine shall be furnished. Any additional data plates shall be as specified and fully described by the procuring activity (see 6.1).

2.19.9 Reference documents. Unless otherwise specified (see 6.1), the documents referenced in this commercial item description shall be the issues in effect on the date of issuance of the invitation for bids or requests for proposal. These documents form a part of this commercial item description to the extent specified. In the event of a conflict between this document and a document referenced herein, this document shall take preference.

3. REGULATOR REQUIREMENTS

3.1 Recovered materials. The contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent possible.

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4. QUALITY ASSURANCE

4.1 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description, and that the product conforms to the producer's own drawings, specifications, standards. and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be othewise provided for under the provisions of the contract.

4.2 Responsibility of inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract. the contractor may use his own or any other facility suitable for the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the commercial item description where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.3 Quality conformance inspection. When specified (see 6.1), quality conformance Inspection shall be applied to production units offered for acceptance under the contract. Unless otherwise specified (see 6.1), quality conformance shall consist of the examination, the tests, and the inspection. One machine from each lot shall be randomly selected for inspection. A lot shall be defined as the number of machines offered for delivery under each contract. If the machine selected for inspection fails the examination, the test, or the inspection, all machines in the lot shall be inspected. Al 1 machines selected for inspection shall pass the examnation, the tests, and Inspection to be accepted.

4.3.1 Examination. Each machine selected for quality conformance inspection shall be visually examined to determine compliance with all requirements of this commercial item description.

4.3.2 Operational test. Each machine selected shall be connected to the electric power source and to the oxygen and fuel gas supply. All power source of the gas system shall be checked for leaks. Gas pressures shall be set in accordance with the manufacturer's recommendation. The torch shall be lighted and adjusted to obtain a neutral flame. The machine shall be operated to assure conformance with all requirements specified herein without performing any cutting operations.

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4.3.3 Cutting test. When specified (see 6.1), a sample of SAE 1020 low carbon steel 1/4 inch thick and 48 inches long shall be cut. The time for the cut shall be not greater than 2 minutes. The accuracy of the cut shall be as specified in 2.14. The cut edge of the kerf shall be square. When specified (see 6.1), a sample of SAE 1020 low carbon steel 6 inches thick and 12 inches wide shall be cut. The time required for the 12 inch cut shall be not greater than 2 minutes. When specified (see 6.1), a sample of SAE 1020 low carbon steel 1/4 inch thick shall be used to cut the minimum, a 4 inch diameter or the maximum, a 54 inch diameter circle. When specified {see 6.1), both circles may be required. The accuracy of the cuts shall be as specified in 2.14.

4.3.4 Environmental protection certification of conformance. The contractor shall certify and maintain substantiating evidence that the product offered complies with all applicable environmental protection regulations. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

5. PRESERVATION, PACKAGING, PACKING LABELING, AND MARKING

5.1 Preservation, packaging, packing, labeling, and marking. Preservation, packaging, packing, labeling, and marking shall be as specified in the contract or purchase order.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents.

a. Title, number, and date of this commercial item description.

b. Quantity required.

c. If torch is to be used with different type, gas, specify type (see 2.7).

d. Additional tips, of required, describe in detail (see 2.8).

e. Additional lengths of interconnectable track, if required (see 2.10).

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f. Electrical system, if different (see 2.13).

g. Electrical input power, if different (see 2.13).

h. Additional performance requirements, if required (see 2.15).

i. Specify and fully describe optional equipment, if required (see 2.17).

j. Repair parts, if required (see 2.18).

k. Additional safety and health requirements, if required (see 2.19.1).

l. Data plates, if different (see 2.19.8).

m. Data of reference documents, if different (see 2.19.9).

n. Quality conformance if required (see 4.3).

o. Cutting test, if required, specify thickness of material to be cut (see 4.3.3).

6.2 Applicable documents.

Federal Standards, FED-STD-376 and FED-STD-H28 are available from The Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia. PA 19120-5099.

OSHA 29 CFR 1910 is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.

NFPA 79 is available from the National Fire Protection Association Inc., batterymarch Park, Quincy, MA 02269-9101.

UL 123 is available from Underwriters Laboratories, Inc., 333 Pfingsten RD., Northbrook, IL 60062-2096.

IP-7 is available from the Rubber Manufacturers Association, 1400 K Street, N.W., Washington, DC 20005-2403.

CGA E-1 is available from the Compressed Gas Associations, Inc., 500 Fifth Ave., New York, NY 10110.

ASTM E 380 and ANSI/1-EEE 268 is available from the American Society for Testing and Materials, 1916 Race ST., Philadelphia, PA 19103-0001.

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